

## 26 Supersymmetric Field Theories

### 26.8 General Kahler Potentials<sup>1</sup>

There are several circumstances in which it is necessary to consider non-renormalizable Lagrangian densities of the general form (26.3.30)

$$\mathcal{L} = 2 \operatorname{Re}[f(\Phi)]_{\mathcal{F}} + \frac{1}{2}[K(\Phi, \Phi^*)]_D, \quad (26.8.1)$$

where the superpotential  $f$  is an arbitrary function of left-chiral scalar superfields  $\Phi_n$  but not their derivatives, and the Kahler potential  $K$  is an arbitrary function of the  $\Phi_n$  and  $\Phi_n^*$  but not their derivatives.

This situation arises in effective field theories whose symmetries rule out any renormalizable interactions, or in which the renormalizable interactions all happen to be small. It is often then possible to calculate scattering amplitudes at low energy from tree graphs, using a Lagrangian with the smallest value for some combination of the numbers of derivatives, fermion fields, and any small renormalizable couplings. In Section 19.5 we examined such an effective field theory with no renormalizable couplings, involving nucleons and soft pions. The dynamically broken gauge theories discussed in Section 21.4 provide examples of effective field theories of this sort with small renormalizable couplings. This circumstance also arises in supersymmetric theories whose symmetries do not allow a superpotential, or where the superpotential is for some reason small. We will encounter an example of this sort when we consider the extended  $N = 2$  supersymmetric theory of Abelian gauge superfields and gauge-neutral chiral scalar superfields in Section 29.5. We will show there that low-energy scattering amplitudes in this theory are generated by tree graphs using a Lagrangian density of the form (26.8.1), with  $f = 0$  and  $K$  a function only of the  $\Phi_n$  and  $\Phi_n^*$  but no derivatives, plus  $\mathcal{F}$ -terms quadratic in gauge superfields. The inclusion of Kahler potentials having arbitrary dependence on  $\Phi_n$  and  $\Phi_n^*$  but not on their derivatives is particularly important in effective field theories in which some scalar fields are of the same order of magnitude as the fundamental energy scale of the underlying theory, although all other field values and all energies are much smaller. This will be of interest, for instance, in connection with theories of gravity-mediated supersymmetry breaking, discussed in Section 31.6.

Let us consider how to express the Lagrangian density (26.8.1) in terms of component fields. We did not use the assumption that  $f(\Phi)$  is a *cubic* polynomial in deriving Eq. (26.4.4), so this continues to give the  $\mathcal{F}$ -terms in the Lagrangian contributed by an arbitrary superpotential. To derive the  $D$ -terms, we note that the term in the Kahler

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<sup>1</sup>This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

potential of fourth order in  $\theta$  is

$$\begin{aligned}
K(\Phi, \Phi^*)|_{\theta^4} = & -\frac{1}{8}(\bar{\theta}\gamma_5\theta)^2 \sum_n \left[ \frac{\partial K(\phi, \phi^*)}{\partial \phi_n} \square \phi_n + \frac{\partial K(\phi, \phi^*)}{\partial \phi_n^*} \square \phi_n^* \right] \\
& + \sum_{nm} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m^*} (\bar{\theta}\gamma_5\theta) [(\bar{\theta}\psi_{mR})(\bar{\theta}\not{\partial}\psi_{nL}) - (\bar{\theta}\psi_{nL})(\bar{\theta}\not{\partial}\psi_{mR})] \\
& + 2 \operatorname{Re} \sum_{nml} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\theta_L^T \epsilon \psi_{nL})(\theta_L^T \epsilon \psi_{mL})(\theta_L^T \epsilon \theta_L)^* \mathcal{F}_\ell^* \\
& + 2 \operatorname{Re} \sum_{nml} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\theta}\psi_{mL})(\bar{\theta}\psi_{\ell R})(\bar{\theta}\gamma_5\gamma_\mu\theta) \partial^\mu \phi_n \\
& + \sum_{nm\ell k} \frac{\partial^4 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^* \partial \phi_k^*} (\bar{\theta}\psi_{nL})(\bar{\theta}\psi_{mL})(\bar{\theta}\psi_{\ell R})(\bar{\theta}\psi_{kR}) \\
& - \frac{1}{4} \sum_{nm} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m^*} \mathcal{F}_n \mathcal{F}_m^* (\bar{\theta}(1+\gamma_5)\theta) (\bar{\theta}(1-\gamma_5)\theta) \\
& + \frac{1}{4} (\bar{\theta}\gamma_5\gamma^\mu\theta) (\bar{\theta}\gamma_5\gamma^\nu\theta) \sum_{mn} \left[ -\frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m^*} \partial_\mu \phi_n \partial_\nu \phi_m^* \right. \\
& \quad \left. + \frac{1}{2} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m} \partial_\mu \phi_n \partial_\nu \phi_m + \frac{1}{2} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n^* \partial \phi_m^*} \partial_\mu \phi_n^* \partial_\nu \phi_m^* \right].
\end{aligned} \tag{26.8.2}$$

We can again use Eqs. (26.A.18) and (26.A.19) as well as Eq. (26.A.9) to put the  $\theta$  dependence of this expression in the form of an over-all factor  $(\bar{\theta}\gamma_5\theta)^2$ , and find

$$\begin{aligned}
K(\Phi, \Phi^*)|_{\theta^4} = & \frac{1}{4}(\bar{\theta}\gamma_5\theta)^2 \left\{ -\frac{1}{2} \sum_n \frac{\partial K(\phi, \phi^*)}{\partial \phi_n} \square \phi_n - \frac{1}{2} \sum_n \frac{\partial K(\phi, \phi^*)}{\partial \phi_n^*} \square \phi_n^* \right. \\
& + \sum_{nm} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m^*} [(\bar{\psi}_m \not{\partial} \psi_{nL}) + (\bar{\psi}_n \not{\partial} \psi_{mR}) - 2\mathcal{F}_n \mathcal{F}_m^*] \\
& + 2 \operatorname{Re} \sum_{nml} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_n \psi_{mL}) \mathcal{F}_\ell^* \\
& - 2 \operatorname{Re} \sum_{nml} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_m \gamma^\mu \psi_{\ell R}) \partial_\mu \phi_n \\
& - \frac{1}{2} \sum_{nm\ell k} \frac{\partial^4 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^* \partial \phi_k^*} (\bar{\psi}_n \psi_{mL})(\bar{\psi}_k \psi_{\ell R}) \\
& + \sum_{nm} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m^*} \partial_\mu \phi_n \partial^\mu \phi_m^* \\
& - \frac{1}{2} \sum_{nm} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m} \partial_\mu \phi_n \partial^\mu \phi_m \\
& \left. - \frac{1}{2} \sum_{nm} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n^* \partial \phi_m^*} \partial_\mu \phi_n^* \partial^\mu \phi_m^* \right\}.
\end{aligned} \tag{26.8.3}$$

To make the reality properties of the fermion kinematic terms transparent, we can use

Eq. (26.A.21) to write

$$(\bar{\psi}_n \not{\partial} \psi_{mR}) = (\bar{\psi}_n \not{\partial} \psi_{mL})^*.$$

The  $D$ -term of  $K(\Phi, \Phi^*)$  is the coefficient of  $-(\bar{\theta}\gamma_5\theta)^2/4$  minus half the d'Alembertian of the  $\theta$ -independent term in  $K(\Phi, \Phi^*)$ , which is just  $K(\phi, \phi^*)$ , so

$$\begin{aligned} \frac{1}{2}[K(\Phi, \Phi^*)]_D = & \text{Re} \sum_{nm} \mathcal{G}_{nm} \left[ -\frac{1}{2}(\bar{\psi}_m \not{\partial} (1 + \gamma_5) \psi_n) + \mathcal{F}_n \mathcal{F}_m^* - \partial_\mu \phi_n \partial^\mu \phi_m^* \right] \\ & - \text{Re} \sum_{nm\ell} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_n \psi_{mL}) \mathcal{F}_\ell^* \\ & + \text{Re} \sum_{nm\ell} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_m \gamma^\mu \psi_{\ell R}) \partial_\mu \phi_n \\ & + \frac{1}{4} \sum_{nm\ell k} \frac{\partial^4 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^* \partial \phi_k^*} (\bar{\psi}_n \psi_{mL}) (\bar{\psi}_k \psi_{\ell R}), \end{aligned} \quad (26.8.4)$$

where  $\mathcal{G}(\phi, \phi^*)$  is the *Kahler metric*

$$\mathcal{G}_{nm}(\phi, \phi^*) \equiv \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m^*}. \quad (26.8.5)$$

Note that the constant matrix  $g_{nm}$  in Eq. (26.4.2) is replaced here with the Kahler metric  $\mathcal{G}_{nm}(\phi, \phi^*)$ . Because the Kahler metric is field-dependent, we cannot in general make it equal to the unit matrix by a field redefinition, so the total Lagrangian must be left in the form

$$\begin{aligned} \mathcal{L} = & \text{Re} \sum_{nm} \mathcal{G}_{nm} \left[ -\frac{1}{2}(\bar{\psi}_m \not{\partial} (1 + \gamma_5) \psi_n) + \mathcal{F}_n \mathcal{F}_m^* - \partial_\mu \phi_n \partial^\mu \phi_m^* \right] \\ & - \text{Re} \sum_{nm\ell} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_n \psi_{mL}) \mathcal{F}_\ell^* \\ & + \text{Re} \sum_{nm\ell} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_m \gamma^\mu \psi_{\ell R}) \partial_\mu \phi_n \\ & + \frac{1}{4} \sum_{nm\ell k} \frac{\partial^4 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^* \partial \phi_k^*} (\bar{\psi}_n \psi_{mL}) (\bar{\psi}_k \psi_{\ell R}) \\ & - \text{Re} \sum_{nm} \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} (\bar{\psi}_n \psi_{mL}) + 2 \text{Re} \sum_n \mathcal{F}_n \frac{\partial f(\phi)}{\partial \phi_n}. \end{aligned} \quad (26.8.6)$$

The bilinear  $(\bar{\psi}_m \not{\partial} \gamma_5 \psi_n)$  is a total derivative, and could be discarded if  $\mathcal{G}_{nm}$  were a constant, but must be kept for general Kahler potentials. This result will be extended to include gauge superfields at the end of Section 27.4.

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As discussed in Section 19.6, the spontaneous breakdown of a global symmetry group  $G$  to a subgroup  $H$  entails the existence of a set of real massless Goldstone bosons with scalar

fields  $\pi_k$ , for which the term in the Lagrangian with the minimum number of derivatives takes the form

$$\mathcal{L}_{G/H} = - \sum_{k\ell} G_{k\ell}(\pi) \partial_\mu \pi_k \partial^\mu \pi_\ell, \quad (26.8.7)$$

where  $G_{k\ell}(\pi)$  is the metric of the coset space  $G/H$ . (Theories with Lagrangian densities of this general form are known as *non-linear  $\sigma$ -models*.) By writing the complex fields  $\phi_n$  in terms of their real and imaginary parts, the term  $-\sum_{nm} \mathcal{G}_{nm}(\phi, \phi^*) \partial_\mu \phi_n \partial^\mu \phi_m^*$  in the Lagrangian density (26.8.6) can be put in the form (26.8.7), but the converse is not generally true: the condition that a set of real coordinates like the Goldstone boson fields  $\pi_k$  can be interpreted as the real and imaginary parts of a set of complex coordinates like the fields  $\phi_n$  with the metric in these coordinates locally given by Eq. (26.8.5) defines what is called a *Kahler manifold*.<sup>2</sup> But it should not be thought that in the common cases where  $G/H$  is *not* a Kahler manifold, it is impossible for  $G$  to be spontaneously broken to  $H$  while keeping supersymmetry unbroken. What happens in these cases is the appearance of extra massless bosons, which, together with the Goldstone bosons, do form a Kahler manifold.

This is because the superpotential  $f(\phi)$  depends on  $\phi$  but not  $\phi^*$ , so if the whole Lagrangian is invariant under a global symmetry group  $G$  then the superpotential is automatically invariant under a group  $G_{\mathbb{C}}$ , the complexification of  $G$ : if  $G$  consists of transformations  $\exp(i \sum_A \theta_A t_A)$  with generators  $t_A$  and arbitrary real parameters  $\theta_A$ , then  $G_{\mathbb{C}}$  consists of transformations  $\exp(i \sum_A z_A t_A)$  with the same generators and arbitrary complex parameters  $z_A$ . (For instance, if  $G$  is  $U(n)$  then  $G_{\mathbb{C}}$  is  $GL(n, \mathbb{C})$ , the group of all complex non-singular matrices, while if  $G$  is  $SU(n)$  then  $G_{\mathbb{C}}$  is  $SL(n, \mathbb{C})$ , the group of all complex non-singular matrices with unit determinant.) Likewise, if some stationary point  $\phi^{(0)}$  of  $f(\phi)$  is left invariant by some subgroup  $H$  of  $G$ , then it will also be left invariant by a subgroup  $H_{\mathbb{C}}$  of  $G_{\mathbb{C}}$ , the complexification of  $H$ . Whether or not  $G/H$  is a Kahler manifold, the complexified coset space  $G_{\mathbb{C}}/H_{\mathbb{C}}$  is always a Kahler manifold. This follows because  $G_{\mathbb{C}}/H_{\mathbb{C}}$  is a complex submanifold of the flat complex space of the  $\phi_n$ , which is a Kahler manifold, and it is a theorem that any complex submanifold of a Kahler manifold is a Kahler manifold [9]. If parameterized by the values of  $\phi_n(z) = [\exp(i \sum_A z_A t_A) \phi^{(0)}]_n$ , the Kahler manifold  $G_{\mathbb{C}}/H_{\mathbb{C}}$  has the metric obtained by embedding it in the flat complex space of the  $\phi_n$ , usually taken with line element  $\sum_n d\phi_n d\phi_n^*$ .

It is true that  $G_{\mathbb{C}}$  is not a symmetry of the whole Lagrangian, but the Goldstone bosons associated with the breakdown of  $G_{\mathbb{C}}$  to  $H_{\mathbb{C}}$  are nevertheless exactly massless. This

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<sup>2</sup>The significance of Kahler manifolds in this context was pointed out in an early paper by Zumino [8]. Note that it is not necessary for the metric to be expressible in the form (26.8.5) over the whole manifold with a single Kahler potential  $K(\phi, \phi^*)$ ; it is only necessary that the manifold can be covered in finite overlapping patches in which this holds, with different Kahler potentials in each patch. The simplest example of a Kahler manifold is the flat complex plane, with Kahler potential  $|z|^2$ . As an example of a coset space  $G/H$  that is a Kahler manifold, Zumino gave the case where  $G = GL(p, \mathbb{C}) \cdot GL(p+q, \mathbb{C})$  and  $H = GL(p, \mathbb{C})$ , with  $p$  and  $q$  arbitrary positive-definite integers, and  $GL(N, \mathbb{C})$  the group of complex non-singular  $N \cdot N$  matrices. The coset space  $G/H$  here has complex coordinates  $\phi_n$  that may be taken as the components of a complex  $p \cdot (p+q)$  matrix  $A$ , which under  $G$  and  $H$  undergoes the transformations  $A \rightarrow BAC$  and  $A \rightarrow BA$ , respectively, where  $B$  and  $C$  are square non-singular complex matrices of dimensionality  $p$  and  $p+q$ , respectively. The Kahler potential in this case is simply  $K \propto \ln \det AA^\dagger$ .

is guaranteed by the non-renormalization theorem of Section 27.6, or, more simply, by the result of Section 25.4, that massless spin zero particles must come in *pairs* that are related by supersymmetry transformations and that therefore have the same transformation under any global symmetry group  $G$  that commutes with supersymmetry.